Mathesh S

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Executive Summary

Skilled in CAD modeling, FEA, CFD, and multidisciplinary system design, I bring a strong simulation-driven approach to product engineering. The role requires bridging theoretical physics with functional, manufacturable designs, which aligns with my project experience in automotive, aerospace, and IoT-integrated systems. I am confident I am a strong fit and eager to contribute effectively.

Technical Skills:

- **CAD & 3D Modeling:** SolidWorks (Parametric, Tolerancing), Solid Edge, Fusion 360
- **Simulation & Analysis:** ANSYS Fluent (CFD), ANSYS Static Structural & Steady-State Thermal (FEA; cantilever beam max deformation 0.1925 mm), Kinematic & Dynamic Motion Analysis
- **Programming & Software:** Python, Embedded C, React.js, Three.js
- **Multidisciplinary Tech:** IoT Telemetry, Digital Twin modeling, Embedded Systems (ESP32, NVIDIA Jetson Orin Nano), Relay-Based Automation

Soft Skills:

- Team Leadership (assigning tasks and coordinating team members)
- Technical Communication (explaining designs to teammates and mentors)
- Time Management (balancing multiple projects with college studies)
- Quick Learning (picking up new tools and software for projects)
- Receptiveness to Feedback (seeking input from mentors and teammates)
- Languages: English, Tamil

Key Engineering Projects:

Smart India Hackathon 2026 (AYASK) | Dashboard: ayask.vercel.app **Aug 2026 – Present**
Role: Multidisciplinary Project Lead (PS: SIH26008, Hardware, Smart Automation)

Responsibilities:

- Developed the complete 465-component SolidWorks assembly (95 unique parts, 16 sub-assemblies) for the iron ore conveyor monitoring system, engineering custom parametric parts and sensor mountings while integrating standardized COTS/vendor hardware. Engineered structural mountings and custom enclosures to securely house ESP32 sensor nodes (MPU6050, DS18B20, HX711 load cell).
- Ensured precise mechanical fit, geometric tolerances, and assembly feasibility for the entire hardware setup.

Impact:

- Delivered a functional predictive-maintenance concept combining IoT sensing, Edge-AI and a live 3D digital twin dashboard, with a sub-second fail-safe relay trip.
- Led and coordinated a team of 6 members, assigning tasks across the project.

Project Vyuha (KPIT Sparkle) **2026 – Present**
Role: Mechanical CAD Designer

Responsibilities:

- Architected the full 56-component robot assembly (45 unique parts, 11 sub-assemblies) in SolidWorks, designing custom actuator housings and low-inertia leg linkages while integrating commercial pancake BLDC motors and standard fasteners.
- Engineered actuator housings and low-inertia leg linkages for the digital twin.

Impact:

- Delivered structural models used in the cloud monitoring dashboard; the proximal actuator layout reduced distal limb inertia by over 40%.
- Collaborated with a team of 5 members under faculty mentor guidance.

MSME Idea Hackathon 6.0 (Omni-Node) **2026**
Role: Hardware Architecture Contributor

Responsibilities:

- Developed an AI-Driven Retrofit Ecosystem for High-Mix Low-Volume (HMLV) CNC machines, adding predictive maintenance and a sub-35 ms protective feed-hold.
- Designed the IP65 fanless Aluminum 6061-T6 enclosure and magnetic sensor pod in SolidWorks, and drafted the hardware architecture and system block diagrams.
- Performed ANSYS steady-state thermal FEA on the enclosure: peak temperature of 36.55 °C, a 43.45 °C safety margin.

Impact:

- Formulated a ₹57,000 non-invasive retrofit framework with a projected 3–4 month payback, bridging modern AI software with legacy CNC hardware.
- Shortlisted in the Top 40 (Round 2); collaborated with a team of 4 members.

Offline Voice-Controlled Switch **2025**
Role: Electronics Integration Lead

Responsibilities:

- Designed the offline speech-recognition architecture on the Ai-Thinker VC-02 module, programmed in Embedded C with 8 trained voice commands.
- Integrated a 4-channel relay module (10 A / 250 V AC) to independently control four appliances by voice.

Impact:

- Built a working prototype with under 200 ms response time at 0.5 W, with no cloud or internet dependency.
- Collaborated with a team of 5 members.

Education

Bachelor of Engineering (B.E.), Mechanical Engineering **Aug 2025 – Present**
Sri Krishna College of Engineering and Technology (SKCET), Coimbatore

Awards & Accomplishments:

- Selected for Dassault Systèmes AAKRUTI Design Challenge (2026)
- Team Lead in Smart India Hackathon 2026 (Teams: AYASK & Tarang)
- Selected in department-level screening, Smart India Hackathon 2026 (PS SIH26008)
- Registered for KPIT Sparkle 2027
- Shortlisted in Top 40 (Round 2), MSME Idea Hackathon 6.0 (Omni-Node)
- Over 30 certifications in CAD, CFD, and Engineering Software (see portfolio)